

Fracture (hip)

Revised: November 24, 2025

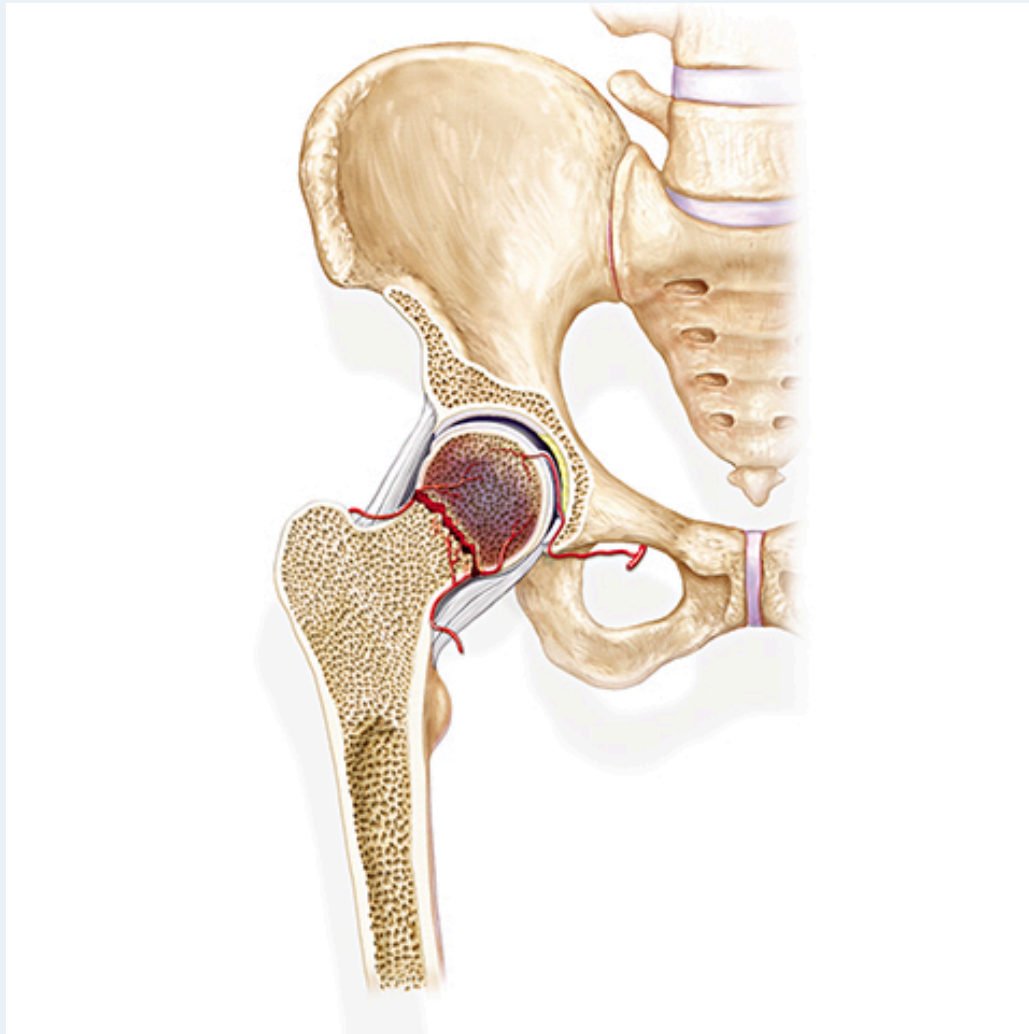
Overview

- Break in the head (most common) or neck of the femur
- Intracapsular (femoral head or neck) or extracapsular (intertrochanteric or subtrochanteric) ^[1] (See [A closer look at hip fracture](#).)
- Most common fall-related injury resulting in hospitalization ^[2]
- Leading cause of disability among older adults
- May permanently change level of functioning and independence
- Fatal in almost 25% of patients within 1 year of fracture ^[3]

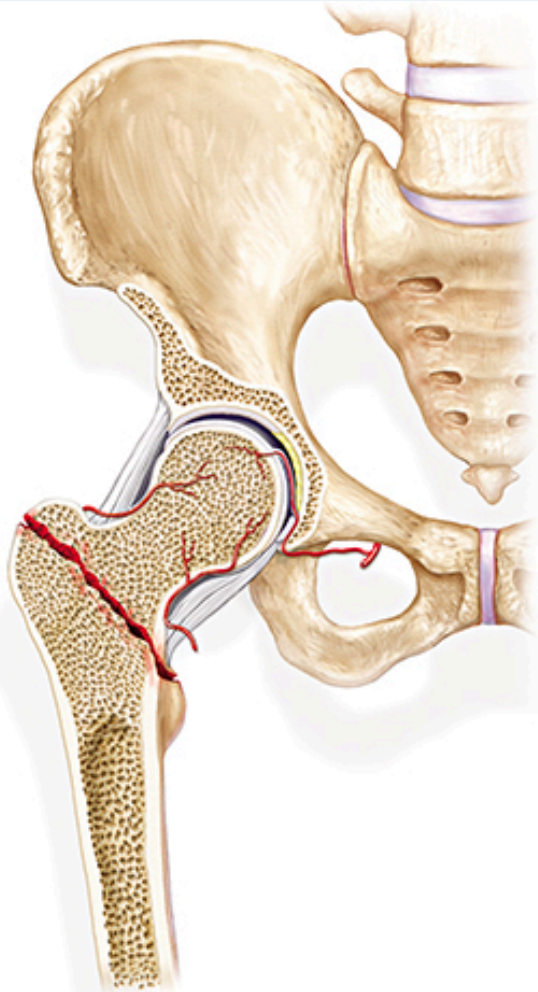
A closer look at hip fracture

The images below show the location of a femoral head fracture and an intertrochanteric fracture.

Femoral head fracture



Intertrochanteric fracture



Pathophysiology

- With a bone fracture, the periosteum and blood vessels in the marrow, cortex, and surrounding soft tissues are disrupted.
- This disruption results in bleeding from the damaged ends of the bone and from the neighboring soft tissue.
- Clot formation occurs within the medullary canal, between the fractured bone ends, and beneath the periosteum.
- Bone tissue immediately adjacent to the fracture dies, and the necrotic tissue causes an intense inflammatory response.
- Vascular tissue invades the fracture area from surrounding soft tissue and marrow cavity within 48 hours, increasing blood flow to the entire bone.
- Bone-forming cells in the periosteum, endosteum, and marrow are activated to produce subperiosteal procallus along the outer surface of the shaft and over the broken ends of the bone.
- Collagen and matrix, which become mineralized to form callus, are synthesized by osteoblasts within the procallus.

- During the repair process, remodeling occurs, unnecessary callus is resorbed, and trabeculae form along stress lines.
- New bone, not scar tissue, forms over the healed fracture.

Causes

- Falls [1]
- Skeletal disease
- Trauma [1]

Risk Factors

- Contact sports participation [1]
- Advanced age [4]
- [Osteoporosis](#) [5]
- Female sex [5] [6]
- Gait problems [5]
- Poor nutrition [5]
- Visual disturbances [5]
- Smoking [7]
- Inactivity [8]
- Chronic medical conditions, such as thyroid or intestinal disorders, that alter vitamin D or calcium absorption or utilization [5]
- Medications that decrease bone density, such as corticosteroids [5]
- Medications that alter cognition or senses (thereby increasing the risk of falls) [5]

Incidence

- Most hip fractures occur in people older than age 65 and in females. [4] [6]
- Femoral head fractures are more common in younger patients; femoral neck fractures, in older adults. [1]
- Intertrochanteric fractures are more common in older adults and in females with [osteoporosis](#). [5]

Complications

- Mental deterioration or [delirium](#)
- Infection
- Avascular necrosis of femoral head
- Phlebitis
- Nonunion

- [Pneumonia](#)
- [Pressure injury](#)
- Social isolation
- [Depression](#)
- Bladder dysfunction
- Thromboembolism
- Hemorrhage
- Hip dislocation
- Death

Assessment

History

- Fall or trauma to the bones
- Pain in the affected hip and leg
- Groin pain
- Pain worsened by movement
- Medications that decrease bone density
- Medications that alter cognition or senses (thereby increasing the risk of falls)

Physical Findings

- Displacement of the extremity
 - Adduction and internal rotation (femoral head fracture)
 - Slight shortening of the extremity as well as abduction and external rotation (femoral neck fracture)
 - Abduction and extension (trochanteric fracture)
 - Significant shortening of the extremity and significant external rotation (intertrochanteric fracture)
- Limited or abnormal range of motion (ROM)
- Edema and discoloration of the surrounding tissue
- In an open fracture, bone protruding through the skin
- Inability to move or bear weight on the affected leg

Diagnostic Test Results

Laboratory

- Blood tests, such as a complete blood count with differential, electrolyte levels, thyroid studies, blood glucose level, and drug and alcohol levels, may identify an underlying cause of the fall or injury.

Imaging

- X-rays show the location of the fracture. [1]
- [Computed tomography scanning \(bone\)](#) or [magnetic resonance imaging](#) shows abnormalities in complicated or occult fractures. [1]
- A [bone scan](#) reveals an occult fracture. [1]

Treatment

General

- Based on age, comorbidities, cognitive functioning, support systems, and functional ability [1] [5]
- Immobilization until repair occurs; possible conservative treatment with rest and protected limited weight-bearing for a compression fracture
- Possible skin traction
- Non-weight-bearing transfers
- Venous thromboembolism (VTE) prophylaxis [9]
- Continuous passive motion devices postoperatively
- Assessment and maintenance of airway, breathing, and circulation if the patient experienced trauma
- Skin care and [pressure injury prevention](#) [10]

Diet

- Well-balanced diet with adequate intake
- Foods rich in vitamins A, C, and D; calcium; and protein
- High-fiber diet if the patient is confined to bed rest
- Nutritional supplements, if indicated

Activity

- Bed rest until repair occurs
- Immobilization of the fracture site
- Ambulation as soon as possible after surgery
- Physical therapy posttreatment [11] [12]

Medications

- Analgesic, such as [acetaminophen](#), [morphine sulfate](#), or an oral opioid [9]
- Prophylactic anticoagulation with [fondaparinux sodium](#), low-dose [heparin sodium](#), low-molecular-weight heparin ([enoxaparin sodium](#), [dalteparin sodium](#)), or [warfarin sodium](#) [9] [13]
- Prophylactic antibiotics, such as [ceFAZolin sodium](#), [tobramycin sulfate](#), and [ampicillin sodium](#)–sulbactam sodium, for patients undergoing immediate internal fixation and those with open fractures [5] [14]
- Bisphosphonate, such as [alendronate sodium](#), [risedronate sodium](#), or [ibandronate sodium](#), to slow bone loss if [osteoporosis](#) is present; often administered with supplemental calcium and vitamin D [5]
- Tranexamic acid to reduce blood loss and blood transfusion in older adult patients with hip fractures [9]

Surgery

- Open reduction and internal fixation using a compression screw and plate or intramedullary nail
- Total hip arthroplasty
- Hemiarthroplasty

Nursing Considerations

Nursing Interventions

- Screen for and assess the patient's pain using facility-defined criteria that are consistent with the patient's age, condition, and ability to understand. [12] [15]
- Treat the patient's pain, as needed and ordered, using nonpharmacologic, pharmacologic, or a combination of approaches. Base the treatment plan on evidence-based practices and the patient's clinical condition, past medical history, and [pain management](#) goals. [12] [15]
- Give prescribed drugs, such as [morphine sulfate](#), initially for pain, and expect to switch to an oral analgesic as soon as possible. [12] [15]
- Monitor closely if the patient is at high risk for adverse outcomes related to opioid treatment, if prescribed. [12] [15]
- Reassess and respond to the patient's pain by evaluating the response to treatment and progress toward [pain management](#) goals. Assess for adverse reactions and risk factors for adverse events that may result from treatment. [12] [15]
- Administer prescribed prophylactic anticoagulation to prevent thrombosis. Use intermittent pneumatic compression devices or antiembolism stockings, as ordered, to prevent VTE.
- Assess the neurovascular status of the affected extremity and compare it with the unaffected extremity.
- Maintain immobilization or traction until surgery, if ordered. Provide care for the patient in traction.
- Maintain proper body alignment and posturing. Assist with quadriceps-setting exercises of the affected extremity to prevent [muscle atrophy](#).
- Use logrolling techniques to turn the patient in bed.
- Maintain non-weight-bearing status.
- Prepare the patient and family for surgery, including what to expect preoperatively and postoperatively.
- Encourage [coughing and diaphragmatic breathing exercises](#) and [incentive spirometry](#) postoperatively.
- Increase the patient's activity level after surgery, as prescribed, and assist with ambulation as soon as possible after surgery. Consult with a physical therapist as early as possible. Institute [safety](#) measures to reduce the risk of injury, such as from falls. Assist with active ROM exercises to

unaffected limbs. [11]

- Institute measures to prevent [pressure injury](#). Perform skin assessments. [10]
- Frequently turn and reposition the patient, unless contraindicated. Base the frequency of repositioning on the patient's tissue tolerance, skin condition, mobility, medical condition, and treatment goals. [16]
- Encourage good nutrition. Offer high-protein, high-calorie snacks and ensure adequate fluid intake.
- Perform daily wound care or surgical site care. [10]
- Obtain specimens for laboratory testing, such as coagulation studies if the patient is receiving anticoagulants.

Monitoring

- Vital signs
- Fluid balance, including intake and output
- Pain level and effectiveness of interventions
- Mobility and ROM
- Incision and dressings
- Complications
- Coagulation study results
- Signs of bleeding
- Neurovascular status
- Skin integrity
- Signs and symptoms of infection

WARNING!



After surgery, assess the patient for signs and symptoms of complications, such as [deep vein thrombosis](#), pulmonary embolus, and hip dislocation.

Associated Nursing Procedures

- [Ambulation, progressive](#)
- [Antiembolism stocking application, knee-length](#)
- [Antiembolism stocking application, thigh-length](#)
- [Cane use training](#)
- [Cold application](#)
- [Coughing and diaphragmatic breathing exercises](#)
- [Fall prevention](#)
- [Heat application](#)
- [Incentive spirometry](#)

- [Internal fixation care and management](#)
- [IV bolus injection](#)
- [IV pump use](#)
- [IV secondary line drug infusion](#)
- [Oral care](#)
- [Oral drug administration](#)
- [Pain assessment](#)
- [Pain management](#)
- [Patient-controlled analgesia](#)
- [Postoperative care](#)
- [Preoperative care](#)
- [Pressure injury prevention](#)
- [Safe medication administration practices, general](#)
- [Sequential compression therapy](#)
- [Surgical wound dressing application](#)
- [Topical skin drug application](#)
- [Transcutaneous electrical nerve stimulation application and use](#)
- [Transcutaneous electrical nerve stimulation removal](#)
- [Walker use training](#)
- [Wound assessment](#)

Patient Teaching

General

Include the patient's family or caregiver in your teaching when appropriate. Provide information according to their individual communication and learning needs. [17] Be sure to cover:

- disorder, diagnostic testing, and treatment, including plans for surgery
- prescribed drugs and potential adverse effects, such as [constipation](#) with opioid analgesics
- [pain management](#) plan and potential adverse effects of [pain management](#) treatment [15]
- safe use, storage, and disposal of opioids, if prescribed [15]
- ROM exercises
- importance of meticulous skin care and proper body alignment
- wound care measures, including postoperative incision site care
- signs and symptoms of infection and complications, such as [deep vein thrombosis](#), and the need to notify a practitioner if any occur
- [coughing and diaphragmatic breathing exercises](#) as well as [incentive spirometry](#)
- use of assistive devices, such as a walker, crutches, or a quad cane
- activity restrictions and lifestyle changes, if appropriate
- safe ambulation practices

- home [safety](#) measures and modifications to reduce the risk of falls and accommodate mobility assistive devices
- exercise program
- importance of a nutritious diet, adequate fluid intake, and intake of high-fiber foods to prevent [constipation](#)
- measures to prevent or reduce the risk of [osteoporosis](#), such as calcium and vitamin D supplementation and the use of a bisphosphonate (such as [alendronate sodium](#), risedronate sodium, or [ibandronate sodium](#))
- importance of follow-up care, including radiographs every 8 to 12 weeks to evaluate healing.

[Discharge Planning](#)

- Participate as part of a multidisciplinary team to coordinate [discharge planning](#) efforts. The team may include a bedside nurse, pharmacist, social worker, case manager, physical therapist, occupational therapist, and orthopedic surgeon. [9] [12]
- Assess the patient's and family's understanding of the diagnosis, treatment, prognosis, follow-up, and warning signs for which to seek medical attention.
- Assess the patient's level of independence before [admission](#).
- Evaluate how the patient's current illness will impact independence.
- Determine the appropriate posthospital setting to which the patient will be discharged.
- Identify the patient's formal and informal supports.
- Identify the patient's and family's goals, preferences, comprehension, and concerns about [discharge](#).
- Confirm arrangements for initial follow-up appointments.
- Provide a list of prescribed drugs, including each drug's dosage, prescribed time schedule, and adverse reactions to report to the practitioner. Provide the patient (and family or caregiver, as needed) with written information on the medications that the patient should take after [discharge](#).
- Assess the patient's and family's understanding of each prescribed medication, including the dosage, administration, expected results, duration, and possible adverse effects.
- Assess the patient's ability to obtain medications. Identify the party responsible for obtaining medications.
- Instruct the patient to provide a list of medications to the practitioner who will be caring for the patient after [discharge](#); to update the information when the practitioner discontinues medications, changes doses, or adds new medications (including over-the-counter products); and to carry a medication list that contains all of this information at all times in the event of an emergency. Remind the patient to discard old medication lists upon returning home. [18] [19]
- Provide information on [smoking cessation](#), if indicated.
- Confirm arrangements for home health care and rehabilitation services, if needed.
- Ensure that the patient and caregivers receive medical contact information.
- Ensure that the patient or caregiver receives a copy of the [discharge](#) instructions and that a copy is placed in the patient's medical record.
- Provide contact information for local support groups and services.
- Document the [discharge planning](#) evaluation in the patient's clinical record, including who was involved in [discharge planning](#) and teaching, their understanding of that teaching, and any need for follow-up teaching.

Related Patient Teaching Handouts

- [Bone density testing](#)
- [Calcium and vitamin D for bone health](#)
- [Femur fracture](#)

- [Femur fracture – ED discharge instructions](#)
- [Fracture in adults – Discharge instructions](#)
- [Fracture in adults – ED discharge instructions](#)
- [Fractures in adults](#)
- [Hip fracture](#)
- [Hip fracture in adults – Discharge instructions](#)
- [How to care for your cast](#)
- [How to care for your child's cast](#)
- [Medicines for osteoporosis](#)
- [Osteoporosis and osteopenia \(low bone mass\)](#)
- [Weight-Bearing Exercises](#)

Resources

- American Academy of Orthopaedic Surgeons: www.aaos.org
- American Orthopaedic Association: <http://aoassn.org>
- Centers for Disease Control and Prevention: www.cdc.gov/falls/prevention
- National Association of Orthopaedic Nurses: <http://orthonurse.org>
- Orthopaedic Trauma Association: <http://ota.org>

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Additional Reading

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([Rating System for the Hierarchy of Evidence for Intervention/Treatment Questions](#))

Rating System for the Hierarchy of Evidence for Intervention/Treatment Questions

The following leveling system is adapted from *Evidence-Based practice in nursing & healthcare: A guide to best practice*, Fifth edition, by Bernadette Mazurek Melnyk and Ellen Fineout-Overholt (2023).

Level I	Evidence from a systematic review or meta-analysis of all relevant randomized controlled trials (RCTs)
Level II	Evidence from well-designed single RCTs (experimental)
Level III	Evidence from well-designed nonrandomized controlled trials (quasi-experimental), systematic reviews of a complete body of evidence, and intervention studies using mixed methods
Level IV	Evidence from well-designed case-control and cohort studies (observational)
Level V	Evidence from systematic reviews of qualitative and descriptive studies
Level VI	Evidence from single descriptive and qualitative studies, evidence-based practice implementation, and quality improvement projects
Level VII	Evidence from expert opinion, expert committee reports, and literature reviews

Data from Gyatt, G., & Rennie D. (2002). *Users' guides to the medical literature*. American Medical Association; Harris, R. P., et al. (2001). *Current methods of the U.S. Preventative Services Task Force: A review of the process*. *American Journal of Preventative Medicine*, 20, 21–35.